WEEK 2 · 16–19 NOVEMBER 2026

Project scoping and use case definition

From an idea to a scope you can actually deliver in three weeks

Week 2 · Session 5 · practical: 16_scene_finder.ipynb

LEARNING OBJECTIVES

What you will be able to do

- Turn a broad idea into a problem statement with a named user and a named decision
- Apply the *why hyperspectral* test, and accept an honest negative answer
- Check that suitable imagery exists **before** committing to an area of interest
- Draft a validation plan before building anything
- Fill in the scoping canvas and name your risks and fallbacks

1. What makes a good scope

Narrow, useful, and provable

THE TEST

A scope is a promise you can keep in three weeks.

Problem → user → decision → product. If you cannot name the person who acts on your output and the decision they take, you have a demo, not a project.

The four questions every scope must answer

- **Problem** — who has what problem, where, and how big is it?
- **User** — which organisation and role receives the output?
- **Decision** — what will they do differently because of it?
- **Product** — in what form does it arrive: a map, a number, an alert, a report?
  - A good test: write the sentence *"Every month, <user> opens <product> and decides <decision>"*. If you cannot, narrow the scope

BE HONEST EARLY

The *why hyperspectral* test

- Name the **specific spectral feature** you rely on, with wavelengths
- Ask whether Sentinel-2 could resolve it — if yes, hyperspectral is not the reason your idea works
- Notebook 15 measured the honest baseline: for broad land cover, 352 bands beat simulated Sentinel-2 bands by **0.3 points** of accuracy
- Notebooks 13 and 14 show where the spectrum wins decisively: mineral species, crop residue, pigments, gases, sub-pixel targets
- **"Multispectral would do"** is a creditable finding — write it down and pivot the added value

2. Does the data exist?

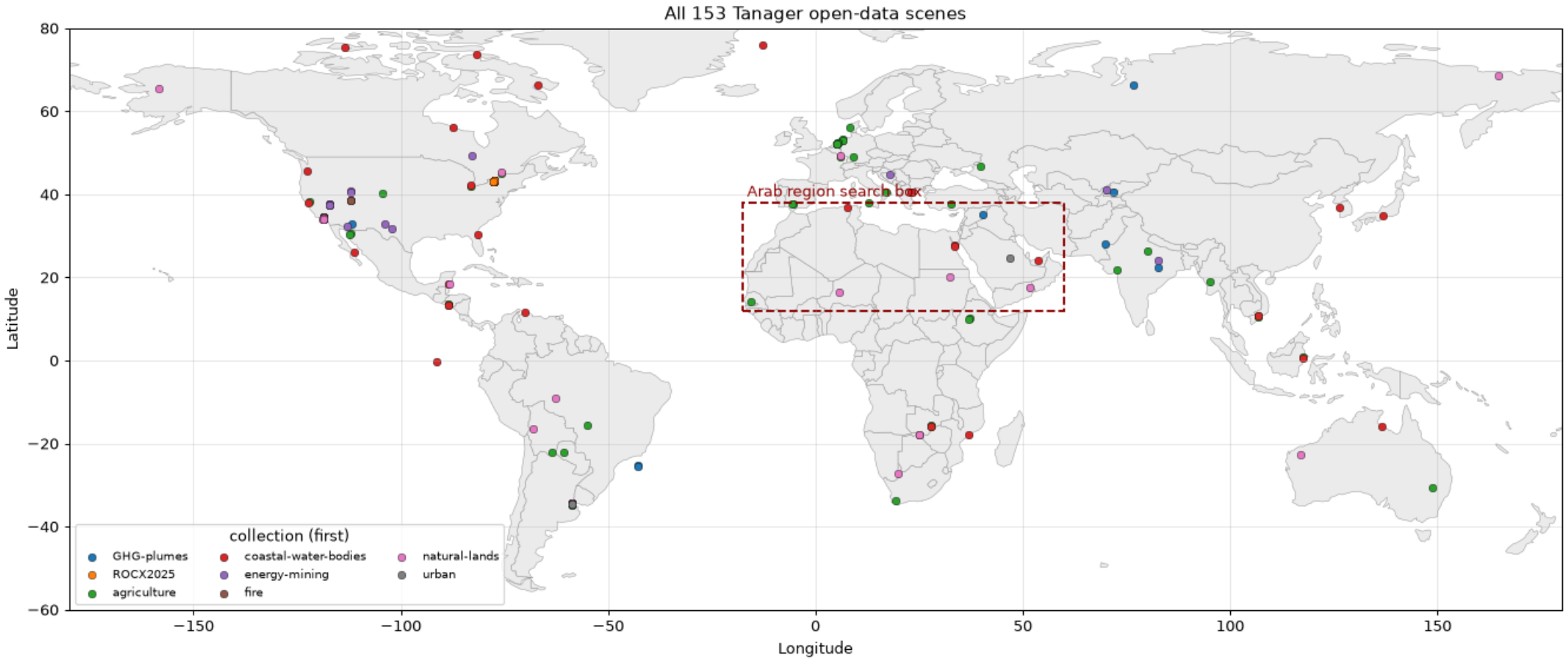
Tasked archives are sparse — check first

Hyperspectral archives are not Sentinel-2

- Tanager is **tasked**: it images requested targets, it does not map the globe on a cycle
- The open archive is therefore **sparse and uneven** — most places have no scene at all
- Repeat coverage exists mainly at supersites, which constrains change-detection projects
- Plan around what exists, or pair one hyperspectral scene with a Sentinel-2 time series

GLOBAL COVERAGE

The Tanager open archive, worldwide



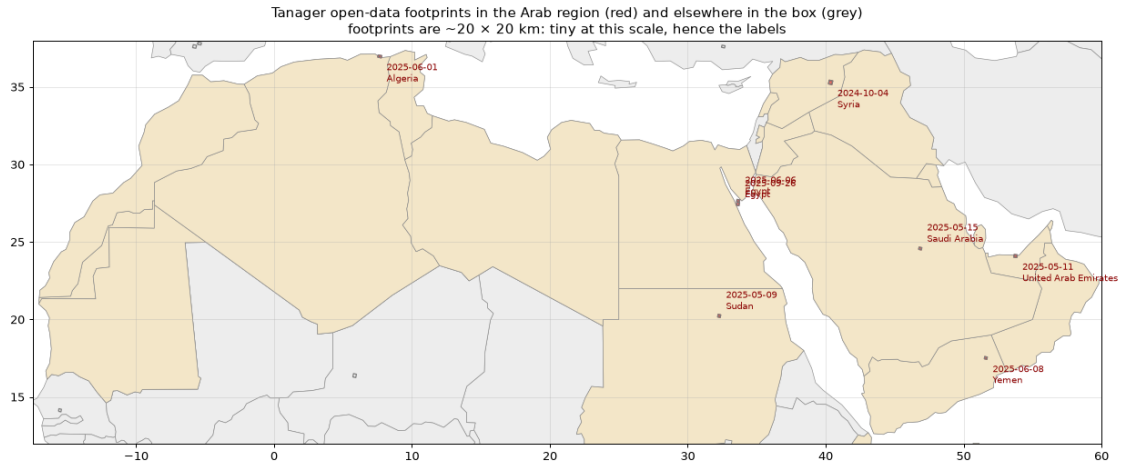
Scene footprints, September 2024 to October 2025

What the open catalogue actually contains

Property	Value
Catalogue entries	276
Unique scenes	153 (123 appear in more than one collection)
Thematic collections	9 (agriculture, urban, fire, GHG-plumes, coastal …)
Date range	25 Sep 2024 – 29 Oct 2025
Scene footprint	322–879 km², median 394 km²
Top countries	United States 52, Argentina 15, Netherlands 9, Canada 7

Searching your own area

- Search on **footprints**, not centroids — a centroid misses partial overlaps
- In the Arab-region box: **14** catalogue entries, **8** unique scenes
- Filter on cloud, off-nadir angle and sun elevation, then look at the thumbnails
- Large off-nadir angles mean bigger pixels and different viewing geometry



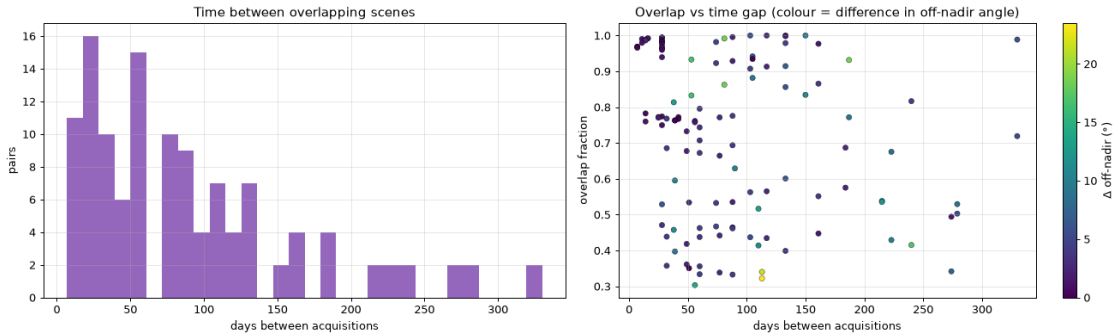
Tanager scenes intersecting the Arab region

LOOK BEFORE YOU TRUST THE METADATA

Three of the eight Arab-region scenes are flagged 50–74 % cloudy.
All three thumbnails are clear.

The provisional cloud mask confuses bright desert sand, gravel and sabkha with cloud: Riyadh 74 %, Al Mahrah 53 %, Al Dhafra 51 %. A strict max-cloud filter would have discarded three of your eight candidate scenes.

Repeat coverage: can you do change detection?



Scenes with repeat coverage in the open archive

- Find **pairs** of scenes whose footprints overlap your area
- Malibu has 6 scenes within ±3 km — one of the few dense supersites
- Check the **time gap** and the **viewing geometry** of each pair
- No pair over your AOI? Use Sentinel-2 for the time axis and Tanager for the detail

Complementary data you should plan for

- **EnMAP** (224 bands, 30 m) — free via DLR registration; delivery takes a few days, so request early
- **Sentinel-2 / Landsat** — the time axis your hyperspectral scene lacks
- **Reference data** for validation: field data, VHR interpretation, ESA WorldCover, national maps
- **Ancillary layers**: DEM, weather, infrastructure, administrative boundaries
  - Register for EnMAP **this week** if your project needs it — it is not an instant download

3. Plan the proof before the build

Validation, risks, milestones

Write the validation plan before you write the model

- Name the **reference data** and how it is independent of your training data
- Choose the **metrics** now: per-class F1 and a confusion matrix, or RMSE and bias, or detection rate at a fixed false-alarm rate
- Decide the **split**: spatial blocks, separate sites, or separate dates
- State a **success criterion** in advance, e.g. *F1 ≥ 0.8 for the mangrove class*
- If you cannot describe how you would be proven wrong, the project is not yet a project

SECTION 10 OF THE CANVAS

Name your risks and your fallbacks now

Risk	Fallback
No cloud-free scene over the AOI	Use another scene from the same collection, or another AOI
Reference data never arrives	Visual interpretation of VHR imagery, or cross-sensor consistency
Classes are not separable	Merge classes, add SWIR features, or switch to unmixing
EnMAP order is delayed	Use the open EnMAP example scenes, or Tanager only
Model does not beat the baseline	Report it honestly; the analysis is still a result

Milestones between here and the final presentation

When	Deliverable
Today, Thu 19 Nov	Scoping canvas drafted, scenes identified and downloading
Mon 23 Nov	Scoping canvas due
Tue 24 Nov	Method applied to your own scene
Wed 25 Nov, pm	Build afternoon — validated result, instructors on hand
Thu 26 Nov	Submission in the morning, presentation in the afternoon

A worked example, in the canvas's own words

- **Problem:** coastal sabkha and shallow-water habitats along the Al Dhafra coast are not mapped at habitat level
- **User and decision:** an environment agency deciding where to survey and protect
- **Why hyperspectral:** gypsum and carbonate absorption features separate sabkha types; narrow visible bands separate shallow benthic cover
- **Data:** one Tanager scene (`20250511_074311_00_4001`, 11 May 2025) — flagged 51 % cloud, confirmed clear by thumbnail
- **Limitation, stated openly:** a single date, so no change analysis; mitigated with Sentinel-2

PRACTICAL

Over to you

▶ `16_scene_finder.ipynb` · `90 min`

- Load the catalogue index and deduplicate to unique scenes
- Search your own area of interest by point or bounding box, with a date window
- Compare cloud metadata with the thumbnails before rejecting any scene
- Look for overlapping pairs if your idea needs change detection
- Fill in section 6 of templates/project_scoping_template.md with what you actually found

KEY TAKEAWAYS

Recap

- Scope = **problem, user, decision, product** — in one sentence you can say out loud
- Apply the *why hyperspectral* test early; "multispectral would do" is a valid, creditable answer
- The archive is **sparse and tasked**: 153 unique scenes, 8 over the whole Arab region
- **Do not trust cloud metadata over deserts** — three of eight Arab-region scenes are false alarms
- Write the **validation plan and the fallbacks** before you write the model

Next: Week 3 — GeoAI methods, then validation and accuracy assessment

References and further reading

Course templates: templates/project_scoping_template.md, templates/evaluation_rubric.md

Course notebooks: 16_scene_finder.ipynb (data availability), 21_validation_accuracy_assessment.ipynb (validation design)

Planet (2024–2025). Tanager-1 open data STAC catalogue — www.planet.com/data/stac

EnMAP data access and open example scenes — www.enmap.org/data_access

Olofsson, P. et al. (2014). Good practices for estimating area and assessing accuracy of land change. RSE 148, 42–57.